

Lecture 6

Filesystem and Management of Data Storages

CoSc2043 – Network and System Administration

Section Overview

- ◆ Devices and Files
- ◆ Filesystem Management
- ◆ Network Filesystems
- ◆ Backups

Kinds of Devices

◆ Physical Devices

- Actual hardware
- Examples
 - ◆ Hard Drives, CDRoms, Floppy
 - ◆ Network cards and modems

◆ Logical Devices

- Pseudo devices
- Examples
 - ◆ Virtual Terminals
 - ◆ Network Ports

Device Drivers and Files

◆ Device drivers

- Software to control hardware
- Compiled into the kernel
- Dynamically loaded module

◆ Device files

- Located in `/dev`
- Provide application access to device
- Driver must also be present to use

Making Device Files

`mknod filename type major minor`

◆ filename: Name of the device file

◆ Type

- "c" – Character Device
- "b" – Block Device

◆ Major Number: Type of device driver

◆ Minor Number: Instance (which one)

◆ `/dev/MAKEDEV`: Creation program

Disk Interfaces

SCSI

Standard interface for servers.

IDE

Standard interface for PCs.

Fibre Channel (FC-AL)

High bandwidth

Can run SCSI or IP

iSCSI

SCSI over fast (e.g., 10-gigabit) IP network equipment.

USB

Fast enough for slow devices on PCs.

SCSI

Small Computer Systems Interface

Fast, reliable, expensive.

A bus, not a simple PC to device interface.

Each device has a target # ranging 0-7 or 0-15.

Devices can communicate directly w/o CPU.

Many versions

Original: SCSI-1 (1979) 5MB/s

Current: SCSI-3 (2001) 320MB/s

Serial Attached SCSI (SAS)

Up to 128 devices

Up to 750 MB/s full duplex.

IDE

Integrated Drive Electronics / AT attachment

Slower, less reliable, cheap.

Only allows 2 devices per interface.

ATAPI standard added removable devices.

Many versions

Original: IDE / ATA (1984)

Current: Ultra-ATA/133 133MB/s

Serial ATA

Up to 128 devices.

150 MB/s (SATA-1) and 300 MB/s (SATA-2)

IDE vs. SCSI

SCSI offers better performance/scale

- Faster bus

- Faster hard drives (up to 15,000rpm).

- Lower CPU usage

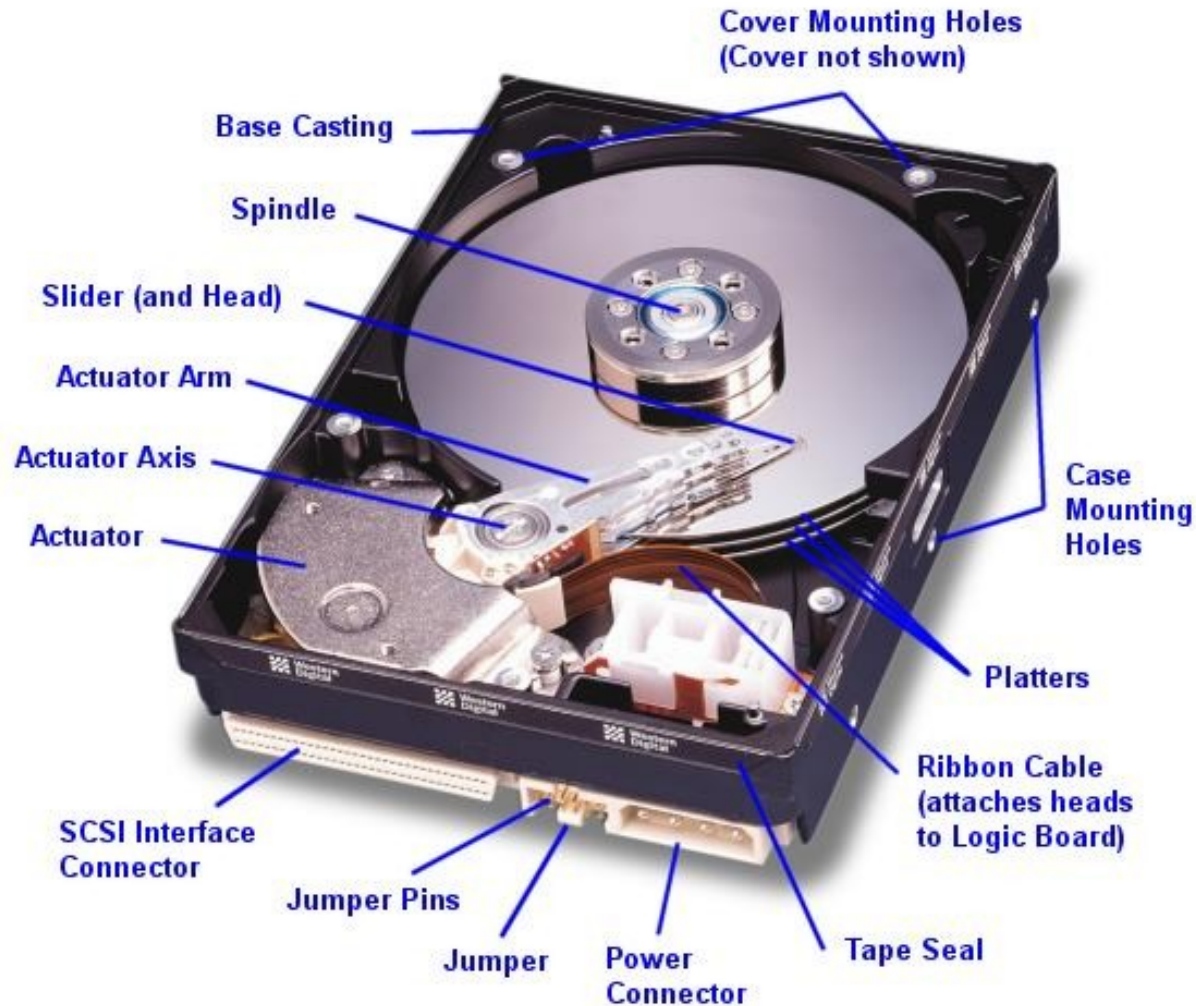
- Better handling of multiple requests.

Cheaper IDE often best for workstations.

Convergence

- SATA2 and SAS converging on a single std.

Hard Drive Components



Hard Drive Components

Actuator

Moves arm across disk to read/write data.

Arm has multiple read/write heads (often 2/platter.)

Platters

Rigid substrate material.

Thin coating of magnetic material stores data.

Coating type determines areal density: Gbits/in²

Spindle Motor

Spins platters from 3600-15,000 rpm.

Speed determines disk latency.

Cache

8-32MB of cache memory

Reliability: write-back vs. write-through

Disk Information: hdparm

```
# hdparm -i /dev/hde
```

```
/dev/hde:
```

```
Model=WDC WD1200JB-00CRA1, FwRev=17.07W17, SerialNo=WD-  
WMA8C4533667
```

```
Config={ HardSect NotMFM HdSw>15uSec SpinMotCtl Fixed DTR>5Mbs  
FmtGapReq }
```

```
RawCHS=16383/16/63, TrkSize=57600, SectSize=600, ECCbytes=40
```

```
BuffType=DualPortCache, BuffSize=8192kB, MaxMultSect=16, MultSect=off
```

```
CurCHS=16383/16/63, CurSects=16514064, LBA=yes, LBASects=234441648
```

```
IORDY=on/off, tPIO={min:120,w/IO:120}, tDMA={min:120,rec:120}
```

```
PIO modes: pio0 pio1 pio2 pio3 pio4
```

```
DMA modes: mdma0 mdma1 mdma2
```

```
UDMA modes: udma0 udma1 udma2 udma3 udma4 *udma5
```

```
AdvancedPM=no WriteCache=enabled
```

```
Drive conforms to: device does not report version:
```

* signifies the current active mode

Disk Performance

Seek Time

Time to move head to desired track (3-8 ms)

Rotational Delay

Time until head over desired block (8ms for 7200)

Latency

Seek Time + Rotational Delay

Throughput

Data transfer rate (20-100 MB/s)

Latency vs. Throughput

Which is more important?

Depends on the type of load.

Sequential access – Throughput

Multimedia on a single user PC

Random access – Latency

Most servers

How to improve performance

Faster disks (15 krpm vs 7200rpm)

Caching (disk, controller, server OS, client OS)

More spindles (disks).

More disk controllers.

Disk Performance: hdparm

```
# hdparm -tT /dev/sda
```

```
/dev/sda:
```

```
Timing cached reads:      1954 MB in 2.00  
seconds = 977.02 MB/sec
```

```
Timing buffered disk reads:
```

```
268 MB in 3.02 seconds = 88.66 MB/sec
```

Reliability

MTBF

Average time between failures ($>100,000$ hours).

Real failure curves

Early phase: high failure rate from defects.

Constant failure rate phase: MTBF valid.

Wearout phase: high failure rate from wear.

Failures more likely on traumatic events.

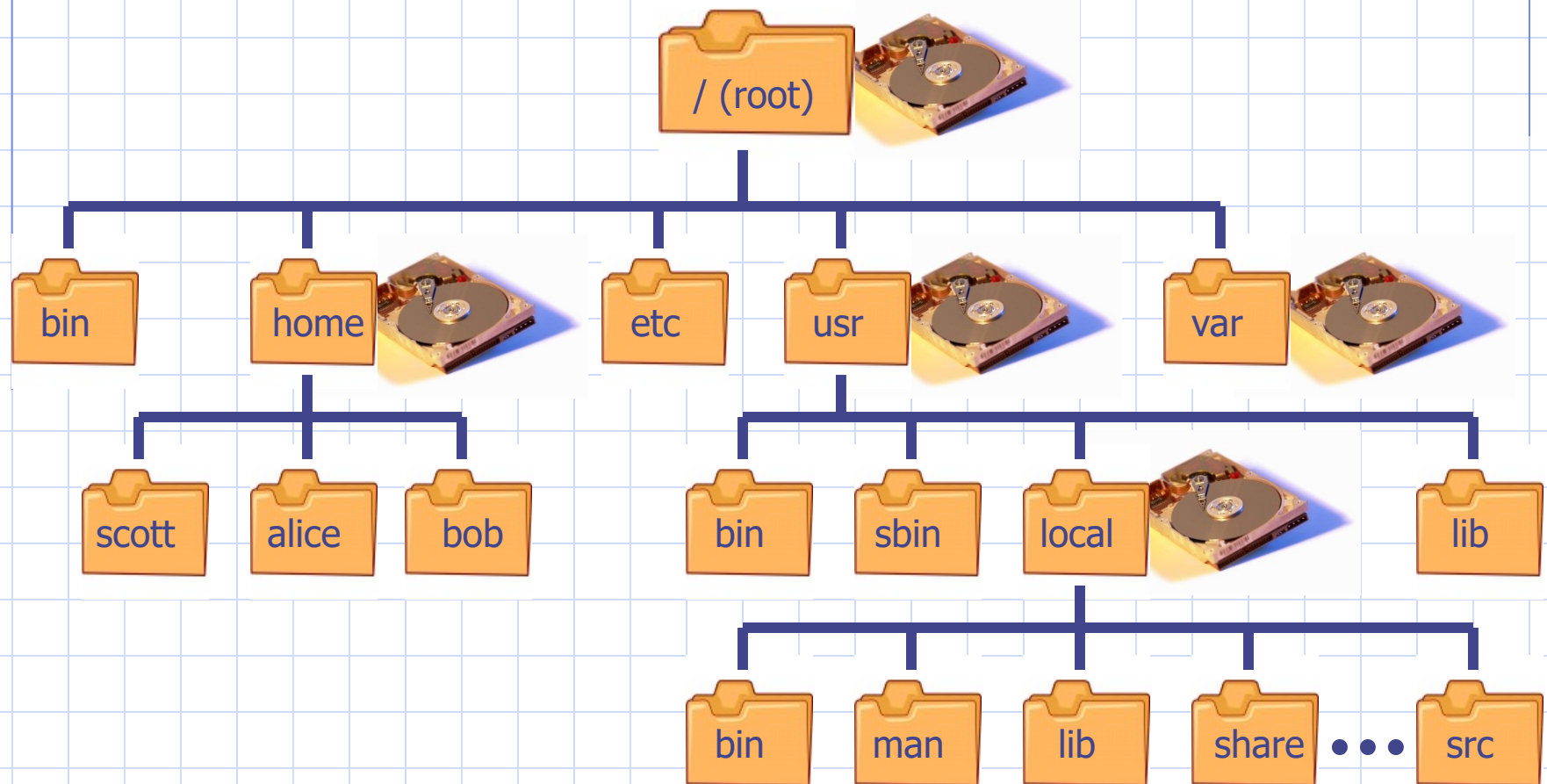
Power on/off.

Systems often wear out before MTBF.

UNIX Filesystems

- ◆ Defines how data is stored on drives
- ◆ Multiple types of filesystems supported
- ◆ Boot Block (1st block)
 - Stores the bootstrap loader program
- ◆ Superblock (2nd block)
 - Copies scattered throughout the partition
 - Stores information about the partition
 - ◆ Partition Size
 - ◆ Type of File system
 - ◆ Block Size
 - ◆ Size & Location of inode tables
 - ◆ Free Block list

Filesystems and Partitions



Why Partition?

- ◆ Not enough space on one disk
- ◆ Separation of data
 - Read-only areas
 - Spool areas and free space
 - OS upgrades
- ◆ Backups
- ◆ Performance

RAID

Redundant Array of Independent Disks

Combine physical disks into single logical unit.

Can be implemented in hardware or software.

Hardware RAID controllers:

- Caching

- Automate rebuilding of arrays

Advantages

- Capacity

- Reliability

- Throughput

RAID Levels

Level	Min	Description
JBOD	2	Merge disks for capacity, no striping.
RAID 0	2	Striped for performance + capacity.
RAID 1	2	Mirrored for fault tolerance.
RAID 3	3	Striped set with dedicated parity disk.
RAID 4	3	Block instead of byte level striping.
RAID 5	3	Striped set with distributed parity.
RAID 6	4	Striped set with dual distributed parity.

Striping

- ◆ Distribute data across multiple disks.
- ◆ Improve I/O by accessing disks in parallel.
 - Independent requests can be serviced in parallel by separate disks.
 - Single multi-block requests can be serviced by multiple disks.
- ◆ Performance vs. reliability
 - Performance increases with # disks.
 - Reliability *decreases* with # disks.

Managing Filesystems

- ◆ Creating a new filesystem
 - Similar to DOS format
 - `mkfs` – creates a filesystem (many aliases)
- ◆ `mount` – mount a partition
 - Mount point must exist (directory)
 - `/etc/fstab`
- ◆ `umount` – Unmount a partition
- ◆ `fsck`
 - Check/fix filesystem errors
 - Journaling filesystems
- ◆ Quotas?

Network Filesystems

- ◆ Filesystems shared via NFS
 - ◆ Client-side: similar to local `mount`
 - ◆ Server – exports shared filesystems
 - `mountd` and `nfsd`
 - `/etc/exports` and `exportfs`
 - Access permissions should be selective
 - ◆ `showmount -e`
 - ◆ Samba – Microsoft filesystem sharing

Why Backups?

- ◆ Hardware failures
- ◆ Accidental deletions or modification
- ◆ Security incidents
- ◆ Upgrades and Migrations

Backup Plan Characteristics

◆ Ease of use

- Automation of backups
- Selective file/directory restores

◆ Time scheduling

◆ Backup verification

◆ Offsite copies

◆ Portability

Backup Media

- ◆ Floppy Disks
- ◆ CD-R and CD-RW Drives
- ◆ Removal Hard drives and Zipdrives
- ◆ Tape
 - 4mm DAT
 - 8mm
 - DLT

Backup tools

- ◆ **dump and restore**
 - filesystem backups
 - Support for incremental backups
- ◆ **tar**
 - File by file backups (archives)
 - Easy to recover selected files
- ◆ **dd** – Duplicate “raw” devices
- ◆ **mt** – Control tape devices
- ◆ **Compression tools**
 - **compress**
 - **gzip**

Backup Strategies

- ◆ Full Backups – Backup entire system
- ◆ Partial Backups – Selective backup
- ◆ Incremental – Backup modified files
- ◆ Basic plan
 - Full backup on Sundays
 - Incremental daily